

SPECIAL SUMMER INTEGRATED LEARNING PROGRAMME MATHEMATICS

Trigonometric Functions

Week 3 Day 2 AM



SINGAPORE UNIVERSITY OF
TECHNOLOGY AND DESIGN

OUTLINE

① Graphs of Sine and Cosine

② More Trigonometric Graphs

③ Inverse Trigonometric Functions and Their Graphs

④ Summary

PERIODICITY OF SINE AND COSINE

To help us to graph $y = \sin x$ and $y = \cos x$, we first observe that they repeat their values when x increases by 2π (when we rotate the angle one full round), i.e.

$$\sin(x + 2\pi) = \sin x \quad \text{and} \quad \cos(x + 2\pi) = \cos x.$$

This is a periodic property, as they repeat their values in a periodic manner. In general, we have the following definition.

DEFINITION:

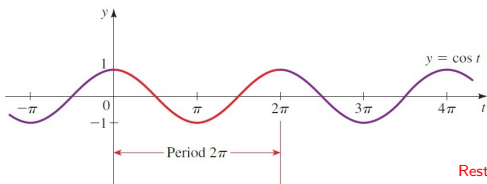
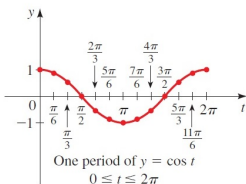
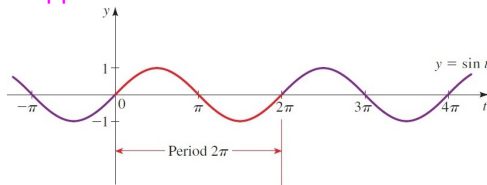
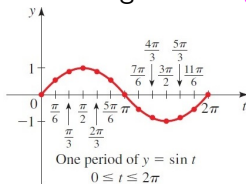
A function f is **periodic** if there exists a positive number p such that $f(x + p) = f(x)$ for all $x \in \mathbb{R}$. The smallest such p (if it exists) is called the **period** of f .

Remark: A nice property of periodic functions is that their graphs is just a repetition of the part over the interval $[0, p]$, where p is the period.

Hence, $\sin x$ and $\cos x$ have period 2π and to sketch their graphs, it's enough to sketch and repeat the part over $[0, 2\pi]$.

GRAPHS OF SINE AND COSINE

Recall that $\sin \theta$ is the y -coordinate of the point on the unit circle determined by an arc length $t = \theta$ measured anti-clockwise from the point $(1, 0)$. So when graphing the function $\sin \theta$, we just need to keep track of the y -coordinate of that point while increasing $t = \theta$. Similarly for $\cos \theta$, it's the x -coordinate instead. Note that the **domains** for $\sin x$ and $\cos x$ are both $\mathbb{R}$. You can visualize these using this [Geogebra applet](#).



GRAPHS OF TRANSFORMATIONS OF SINE AND COSINE

To sketch the graphs of transformations of $\sin \theta$ and $\cos \theta$, simply apply what you have learned in Week 1, no new concepts are involved. We summarize their properties as follows:

DEFINITION:

The graphs of

$$y = a \sin(k(x - b)) \quad \text{and} \quad y = a \cos(k(x - b)) \quad (k > 0)$$

has amplitude $|a|$, period $\frac{2\pi}{k}$, and horizontal shift by b . The **amplitude** of a sinusoidal curve is half of the difference between the maximum and minimum y -values of the curve.

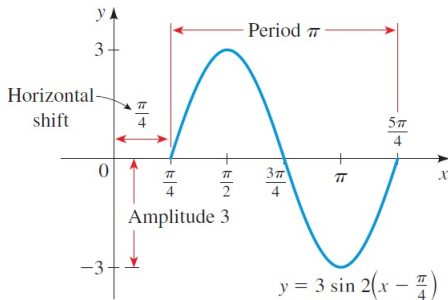
Remarks:

- Note that the period is scaled by $\frac{1}{k}$, which is exactly the horizontal stretching and shrinking.
- If $b > 0$, the horizontal shift is to the right, if $b < 0$, the horizontal shift is to the left.

GRAPHS OF TRANSFORMATIONS OF SINE AND COSINE

- EXAMPLE

E.g.: To graph $y = 3 \sin \left(2x - \frac{\pi}{2} \right) = 3 \sin \left(2 \left(x - \frac{\pi}{4} \right) \right)$, we first shrink the period 2π of the graph of $y = \sin x$ by $\frac{1}{2}$, then shift it to the right by $\frac{\pi}{4}$, and stretch the amplitude to 3.



After that, just repeat the graph in both directions.

ACTIVITY 1

Graph the following functions.

(a) $y = -4 \sin(2x + \pi).$

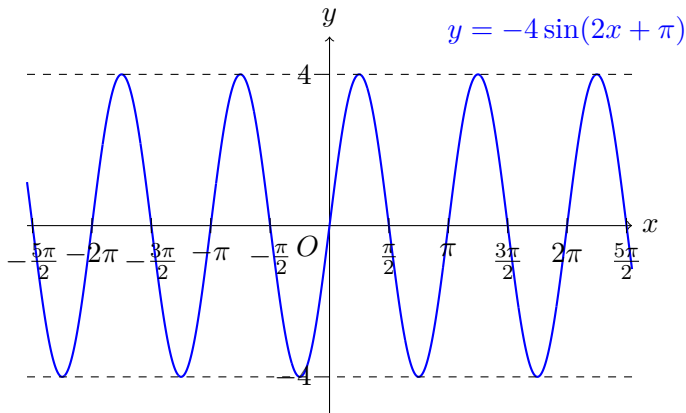
(b) $y = 3 - 4 \sin(2x + \pi).$

(c) $y = \cos\left(3x + \frac{\pi}{2}\right).$

(d) $y = 1 + \cos\left(3x + \frac{\pi}{2}\right).$

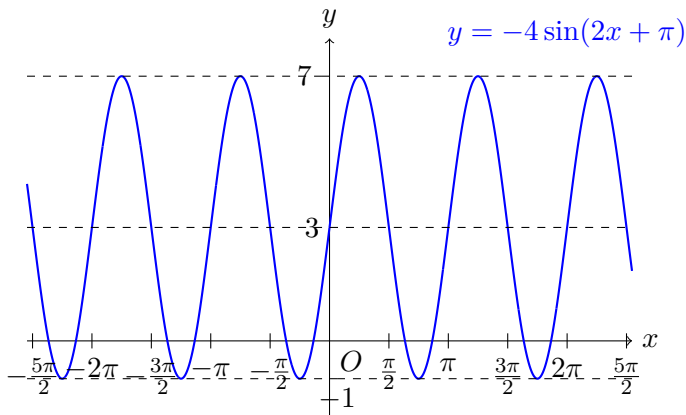
ACTIVITY 1: SOLUTION

- (a) To graph $y = -4\sin(2x + \pi) = -4\sin(2(x - (-\frac{\pi}{2})))$, we first shrink the period 2π of the graph $y = \sin x$ by $\frac{1}{2}$, then shift the graph to the left by $\frac{\pi}{2}$, stretch the amplitude to 4, and reflect it w.r.t. the x -axis (due to the $-$ sign of -4).



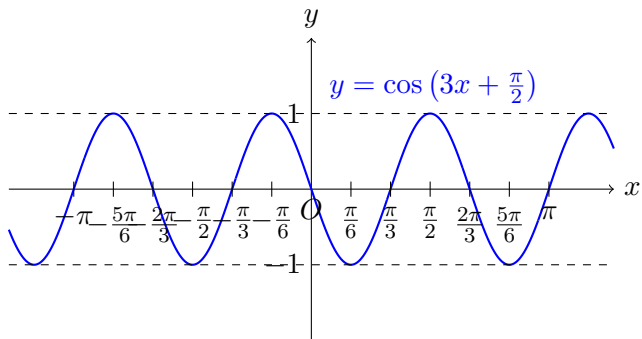
ACTIVITY 1: SOLUTION

- (b) The graph $y = 3 - 4\sin(2x + \pi)$ is obtained by vertically shifting upward the previous graph by 3.



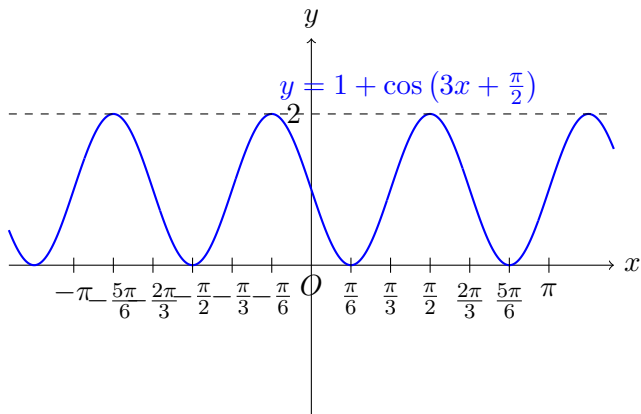
ACTIVITY 1: SOLUTION

- (c) To graph $y = \cos\left(3x + \frac{\pi}{2}\right) = \cos\left(3\left(x - \left(-\frac{\pi}{6}\right)\right)\right)$, we first shrink the period 2π of the graph $y = \cos x$ by $\frac{1}{3}$ and then shift the graph to the left by $\frac{\pi}{6}$.



ACTIVITY 1: SOLUTION

- (d) The graph $y = 1 + \cos\left(3x + \frac{\pi}{2}\right)$ is obtained by vertically shifting upward the previous graph by 1.



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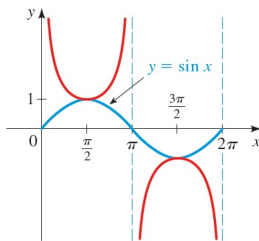
GRAPHS OF SECANT AND COSECANT

Since

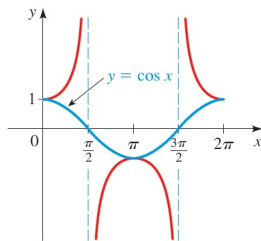
$$\csc x = \frac{1}{\sin x} \quad \text{and} \quad \sec x = \frac{1}{\cos x},$$

the trigonometric functions $\csc$ and $\sec$ have period 2π as well.

Moreover, by using the reciprocal relationship, whenever $\sin x$ is increasing, $\csc x$ is decreasing, whenever $\sin x$ is decreasing, $\csc x$ is increasing. Also, when $\sin x \rightarrow 0^+$, $\csc x \rightarrow \infty$, because we are dividing by a very small positive number; when $\sin x \rightarrow 0^-$, $\csc x \rightarrow -\infty$, because we are dividing by a very small negative number. Similar observations for $\cos x$ and $\sec x$.

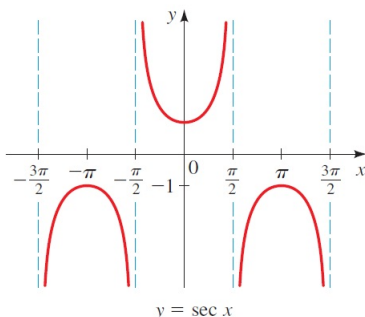
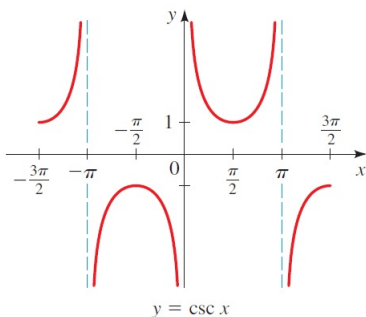


One period of $y = \csc x$



One period of $y = \sec x$

GRAPHS OF SECANT AND COSECANT

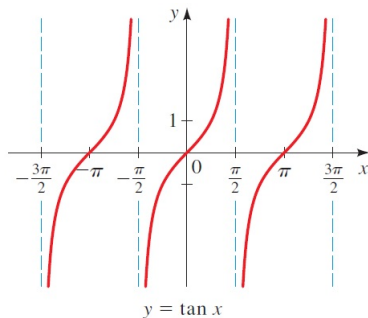


Thus, the lines $x = n\pi$ for $n \in \mathbb{Z}$ (multiples of π) are vertical asymptotes of $\csc x$. The domain of $\csc x$ is $\mathbb{R} \setminus \{n\pi \mid n \in \mathbb{Z}\}$.

Whereas the lines $x = (2n + 1)\frac{\pi}{2}$ for $n \in \mathbb{Z}$ (odd multiples of $\frac{\pi}{2}$) are vertical asymptotes of $\sec x$. The domain of $\sec x$ is $\mathbb{R} \setminus \{(2n + 1)\frac{\pi}{2} \mid n \in \mathbb{Z}\}$.

GRAPHS OF TANGENT

Since $\tan x = \frac{\text{opposite}}{\text{adjacent}}$ of a right triangle, if we fix the adjacent side at length 1, then $\tan x$ is just the length of the opposite side, i.e. the height. You can visualize this using this [Geogebra applet](#).



We see that $y = \tan x$ has period π and vertical asymptotes $x = (2n + 1)\frac{\pi}{2}$ for $n \in \mathbb{Z}$ (odd multiples of $\frac{\pi}{2}$). The domain of $\tan x$ is $\mathbb{R} \setminus \{(2n + 1)\frac{\pi}{2} \mid n \in \mathbb{Z}\}$.

GRAPHS OF COTANGENT

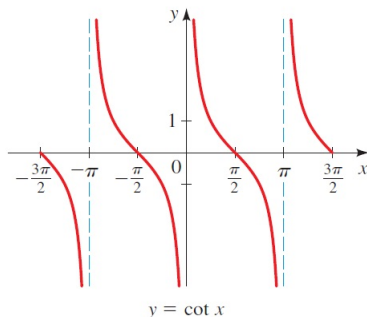
Since $\cot x = \frac{1}{\tan x}$, it has period π as well.

Moreover, by using the reciprocal relationship, whenever $\tan x$ is increasing, $\cot x$ is decreasing, whenever $\tan x$ is decreasing, $\cot x$ is increasing.

Also, when $\tan x \rightarrow 0^+$, $\cot x \rightarrow \infty$, because we are dividing by a very small positive number; when $\tan x \rightarrow 0^-$, $\cot x \rightarrow -\infty$, because we are dividing by a very small negative number.

On the other hand, when $\tan x \rightarrow \infty$, $\cot x \rightarrow 0^+$, because we are dividing by a very huge positive number; when $\tan x \rightarrow -\infty$, $\cot x \rightarrow 0^-$, because we are dividing by a very huge negative number.

GRAPHS OF COTANGENT



We see that the lines $x = n\pi$ for $n \in \mathbb{Z}$ (multiples of π) are vertical asymptotes of $\cot x$. The domain of $\cot x$ is $\mathbb{R} \setminus \{n\pi \mid n \in \mathbb{Z}\}$.

ACTIVITY 2

Apply transformations of functions that you learned in Week 1 to graph the following functions.

(a) $y = 4 \tan(4x - 2\pi)$

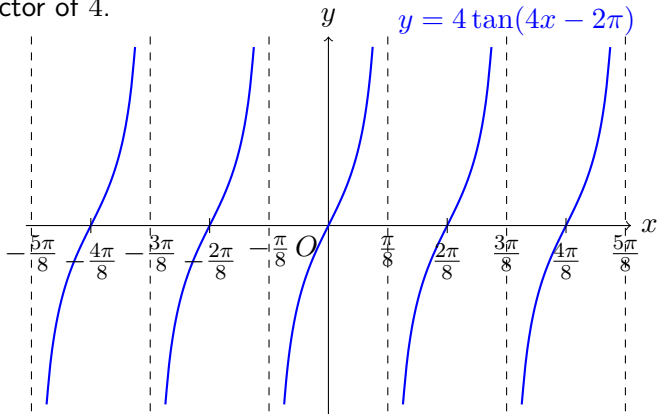
(b) $y = 4 \tan(4x - 2\pi) - 3$

(c) $y = 3 \sec\left(\frac{x}{4} - \frac{\pi}{6}\right)$

(d) $y = 1 + 3 \sec\left(\frac{x}{4} - \frac{\pi}{6}\right)$

ACTIVITY 2: SOLUTION

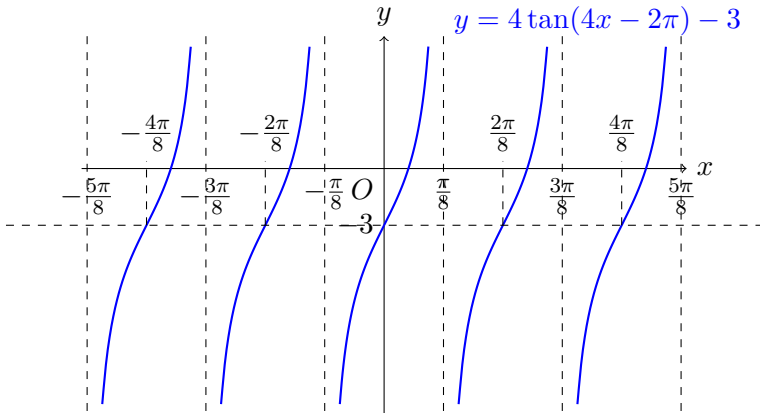
- (a) To graph $y = 4 \tan(4x - 2\pi) = 4 \tan\left(4\left(x - \frac{\pi}{2}\right)\right)$, we first shrink the period π of the graph $y = \tan x$ by $\frac{1}{4}$, then shift the graph to the right by $\frac{\pi}{2}$, and stretch in the y -direction by a factor of 4.



Remark: Since $\tan x$ has period π , we could first simplify $y = 4 \tan(4x - 2\pi) = 4 \tan(4x)$ and just do the two scalings.

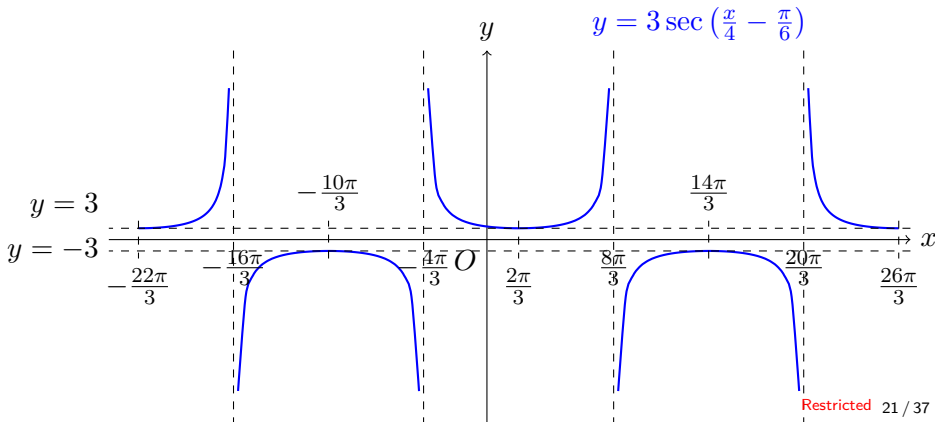
ACTIVITY 2: SOLUTION

- (b) The graph $y = 4 \tan(4x - 2\pi) - 3$ is obtained by vertically shifting downward the previous graph by 3.



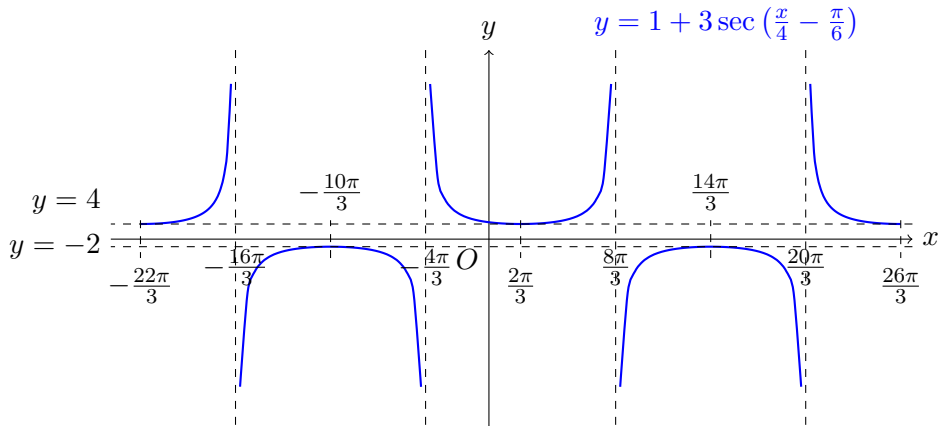
ACTIVITY 2: SOLUTION

- (c) To graph $y = 3 \sec \left(\frac{x}{4} - \frac{\pi}{6} \right) = 3 \sec \left(\frac{1}{4} \left(x - \frac{2\pi}{3} \right) \right)$, we first stretch the period 2π of the graph $y = \sec x$ by a factor of 4, then shift the graph to the right by $\frac{2\pi}{3}$, and stretch in the y -direction by a factor of 3.



ACTIVITY 2: SOLUTION

- (d) The graph $y = 1 + 3 \sec\left(\frac{x}{4} - \frac{\pi}{6}\right)$ is obtained by vertically shifting upward the previous graph by 1



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THE INVERSE SINE FUNCTION

We see easily from the graph that the function $\sin : \mathbb{R} \rightarrow \mathbb{R}$ is not one-to-one and onto. In order to define the inverse sine function, we need to choose a domain and codomain for $\sin$ to define a one-to-one and onto function.

DEFINITION:

The function $\sin : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \rightarrow [-1, 1]$ is one-to-one and onto, thus invertible. Its inverse

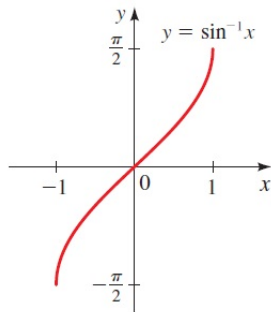
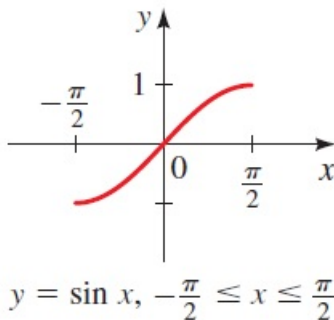
$$\sin^{-1} \text{ or } \arcsin : [-1, 1] \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

is called the **inverse sine function**/**arcsine function**.

⚠ The superscript $^{-1}$ notation follows from the notation of the inverse function. $\sin^{-1} x \neq \frac{1}{\sin x}$. In fact, we've already defined $\frac{1}{\sin x} = \csc x$. To avoid this confusion, you are advised to use $\arcsin$.

THE GRAPH OF ARCSINE FUNCTION

As the inverse function, the graph of $y = \sin^{-1}(x)$ is the reflection of the graph of $y = \sin x$ w.r.t. the line $y = x$.



Note that $\sin^{-1}(x)$ is an odd function, so $\sin^{-1}(-x) = -\sin^{-1}(x)$ for all $x \in [-1, 1]$.

THE INVERSE SINE FUNCTION - EXAMPLE

E.g.: $\sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6}$, because $\sin \frac{\pi}{6} = \frac{1}{2}$ and $\frac{\pi}{6} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

E.g.: $\sin^{-1}\left(-\frac{1}{2}\right) = -\frac{\pi}{6}$, because $\sin\left(-\frac{\pi}{6}\right) = -\frac{1}{2}$ and $-\frac{\pi}{6} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

E.g.: $\sin^{-1}\left(\frac{3}{2}\right)$ is not defined, because $\frac{3}{2}$ is not in the domain $[-1, 1]$ of $\sin^{-1}$.

E.g.: $\sin(\sin^{-1}(1)) = \sin\left(\frac{\pi}{2}\right) = 1$. In general, we always have $\sin(\sin^{-1}x) = x$.

E.g.: $\sin^{-1}(\sin 2\pi) = \sin^{-1}0 = 0$ ($\neq 2\pi$), because $\sin 0 = 0$ and $0 \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

⚠ Due to the restricted domain $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ of sine in order to define an invertible function, the property

$$\sin^{-1}(\sin x) = x$$

is only true for $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

THE INVERSE COSINE FUNCTION

Similarly for the function $\cos : \mathbb{R} \rightarrow \mathbb{R}$, which is not one-to-one and onto, we'll restrict the domain and codomain for $\cos$ in order to define a one-to-one and onto function.

DEFINITION:

The function $\cos : [0, \pi] \rightarrow [-1, 1]$ is one-to-one and onto, thus invertible. Its inverse

$$\cos^{-1} \text{ or } \arccos : [-1, 1] \rightarrow [0, \pi]$$

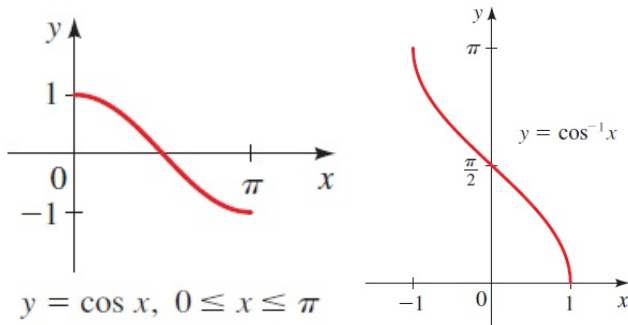
is called the *inverse cosine function* / *arccosine function* .

⚠ The restricted domain of $\cos$ is $[0, \pi]$, which is not equal to the one of $\sin$. In fact, $\cos$ is not one-to-one on $[-\frac{\pi}{2}, \frac{\pi}{2}]$.

⚠ $\cos^{-1} x \neq \frac{1}{\cos x} = \sec x$. To avoid this confusion, you are advised to use $\arccos$.

THE GRAPH OF ARCCOSINE FUNCTION

As the inverse function, the graph of $y = \cos^{-1}(x)$ is the reflection of the graph of $y = \cos x$ w.r.t. the line $y = x$.



⚠ $\cos^{-1}(x)$ is **not** an even function.

THE INVERSE COSINE FUNCTION - EXAMPLE

E.g.: $\cos^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{3}$, because $\cos \frac{\pi}{3} = \frac{1}{2}$ and $\frac{\pi}{3} \in [0, \pi]$.

E.g.: $\cos^{-1}\left(-\frac{1}{2}\right) = \frac{2\pi}{3}$, because $\cos\left(\frac{2\pi}{3}\right) = -\frac{1}{2}$ and $\frac{2\pi}{3} \in [0, \pi]$.

E.g.: $\cos^{-1}\left(\frac{3}{2}\right)$ is not defined, because $\frac{3}{2}$ is not in the domain $[-1, 1]$ of $\cos^{-1}$.

E.g.: $\cos(\cos^{-1}(1)) = \cos(0) = 1$. In general, we always have $\cos(\cos^{-1}x) = x$.

E.g.: $\cos^{-1}(\cos 2\pi) = \cos^{-1}1 = 0$ ($\neq 2\pi$), because $\cos 0 = 1$ and $0 \in [0, \pi]$.

⚠ Due to the restricted domain $[0, \pi]$ of cosine in order to define an invertible function, the property

$$\cos^{-1}(\cos x) = x$$

is only true for $x \in [0, \pi]$.

THE INVERSE TANGENT FUNCTION

We restrict the domain for $\tan$ to $(-\frac{\pi}{2}, \frac{\pi}{2})$ in order to define a one-to-one and onto function.

DEFINITION:

The function $\tan : (-\frac{\pi}{2}, \frac{\pi}{2}) \rightarrow \mathbb{R}$ is one-to-one and onto, thus invertible. Its inverse

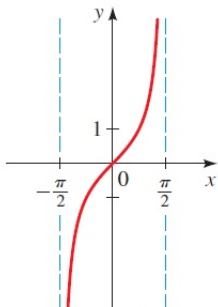
$$\tan^{-1} \text{ or } \arctan : \mathbb{R} \rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

is called the ***inverse tangent function***/***arctangent function***.

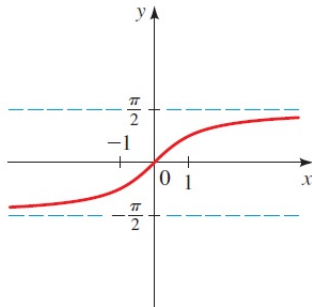
⚠ $\tan^{-1} x \neq \frac{1}{\tan x} = \cot x$. To avoid this confusion, you are advised to use $\arctan$.

THE GRAPH OF ARCTANGENT FUNCTION

As the inverse function, the graph of $y = \tan^{-1}(x)$ is the reflection of the graph of $y = \tan x$ w.r.t. the line $y = x$.



$$y = \tan x, -\frac{\pi}{2} < x < \frac{\pi}{2}$$



$$y = \tan^{-1} x$$

Note that $y = \tan^{-1} x$ has two horizontal asymptotes $y = -\frac{\pi}{2}$ and $y = \frac{\pi}{2}$. Also, $\tan^{-1}(x)$ is an odd function, so $\tan^{-1}(-x) = -\tan^{-1}(x)$ for all $x \in [-1, 1]$.

THE INVERSE TANGENT FUNCTION - EXAMPLE

E.g.: $\tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$, because $\tan \frac{\pi}{6} = \frac{1}{\sqrt{3}}$ and $\frac{\pi}{6} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

E.g.: $\tan^{-1}\left(-\frac{1}{\sqrt{3}}\right) = -\frac{\pi}{6}$, because $\tan\left(-\frac{\pi}{6}\right) = -\frac{1}{\sqrt{3}}$ and $-\frac{\pi}{6} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

E.g.: $\tan^{-1}(\sqrt{3})$ is defined, as the domain of $\tan^{-1}$ is $\mathbb{R}$. Since $\tan\left(\frac{\pi}{3}\right) = \sqrt{3}$ and $\frac{\pi}{3} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, so $\tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$.

E.g.: $\tan(\tan^{-1}(1)) = \tan\left(\frac{\pi}{4}\right) = 1$. In general, we always have $\tan(\tan^{-1}x) = x$.

E.g.: $\tan^{-1}(\tan 2\pi) = \tan^{-1}0 = 0$ ($\neq 2\pi$), because $\tan 0 = 0$ and $0 \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

⚠ Due to the restricted domain $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ of tangent in order to define an invertible function, the property

$$\tan^{-1}(\tan x) = x$$

is only true for $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

ACTIVITY 3

Evaluate the following values.

(a) $\arcsin\left(\frac{\sqrt{2}}{2}\right)$

(b) $\arccos\left(-\frac{\sqrt{2}}{2}\right)$

(c) $\arctan(-1)$

(d) $\sin\left(\sin^{-1}\left(\frac{1}{4}\right)\right)$

(e) $\cos^{-1}\left(\cos\left(\frac{7\pi}{6}\right)\right)$

(f) $\sin\left(\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)\right)$

(g) $\tan\left(\sin^{-1}\left(\frac{3}{5}\right)\right)$

Hint: draw a right triangle.

ACTIVITY 3: SOLUTION

$$(a) \arcsin\left(\frac{\sqrt{2}}{2}\right) = \frac{\pi}{4}.$$

$$(b) \arccos\left(-\frac{\sqrt{2}}{2}\right) = \frac{3\pi}{4}.$$

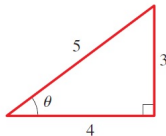
$$(c) \arctan(-1) = -\frac{\pi}{4}.$$

$$(d) \sin\left(\sin^{-1}\left(\frac{1}{4}\right)\right) = \frac{1}{4}.$$

$$(e) \cos^{-1}\left(\cos\left(\frac{7\pi}{6}\right)\right) = \cos^{-1}\left(-\cos\left(\frac{\pi}{6}\right)\right) = \cos^{-1}\left(-\frac{\sqrt{3}}{2}\right) = \frac{5\pi}{6}.$$

$$(f) \sin\left(\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)\right) = \sin\frac{5\pi}{6} = \frac{1}{2}.$$

$$(g) \tan\left(\sin^{-1}\left(\frac{3}{5}\right)\right) = \tan\theta = \frac{3}{4}.$$



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SUMMARY

- Graphs of Sine and Cosine
 - periodicity of sine and cosine.
 - graphs of sine and cosine, graphs of transformations of sine and cosine.
- More Trigonometric Graphs
 - graphs of secant and cosecant.
 - graphs of tangent and cotangent.
- Inverse Trigonometric Functions and Their Graphs
 - the inverse sine function and its graph.
 - the inverse cosine function and its graph.
 - the inverse tangent function and its graph.

RESOURCES

References:

- James Stewart. Precalculus Mathematics for Calculus (7th edition) sections 5.3, 6.4, 5.5.